University of Toronto
Faculty of Arts and Science

MAT137Y Test 2

A. Adames, J. Cruz Chapital , X. Cui, G. Huang, B. Khesin, A. Rakshit, A. Rolf, I. Sierra Del Rio, Y. Wang

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No Aids Permitted

Last Name: Kim

First Name: Minsung

UToronto email: minsungg.kim@mail.utoronto.ca

UTORid: Kimmi396

Student Number: 1012139213

Please make sure your information above are correct.

This exam contains 10 **one-sided pages** (including this cover page) and is printed on 10 sheets of paper. There are 7 problems.

Instructions

- **Keep this booklet closed**, until an invigilator announces that the test has begun. However, you may fill out your information in this page before the test starts.
- Please place your **student ID** in a location on your desk that is easy for an invigilator to check without disturbing you during the test.
- **Turn your cell phone completely off** (not just in vibrate mode) and store it with your other belongs.
- **No aids are permitted on this examination.** Examples of illegal aids include but are not limited to textbooks, notes, calculators, cellphones, or any electronic device.
- If you need **space for your draft work**, you can use the back of each page as your scratch papers. Note that we will not scan the back of each page.
- **Write clearly** and concisely by using **dark pencil or pen.** Work scattered all over the page without a clear ordering will receive very little credit.
- For questions with a boxed area, ensure your answer is completely **inside the box**.
- **If you need more space**, use the blank pages at the end of the exam and clearly indicate on the question page when you have done this. **Do not tear any pages off this exam.**

Question:	1	2	3	4	5	6	7	Total
Points:	4	8	6	7	7	5	7	44
Score:								

Do not tear this page off. This page is a formula sheet and can be used for rough work only. It will not be graded under any circumstances.

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \sec\theta = \frac{1}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta} = \frac{1}{\tan\theta} \qquad \csc\theta = \frac{1}{\sin\theta}$$

$$\sin^2\theta + \cos^2\theta = 1 \qquad 1 + \tan^2\theta = \sec^2\theta \qquad 1 + \cot^2\theta = \csc^2\theta$$

$$\begin{aligned}\cos 2\theta &= \cos^2\theta - \sin^2\theta \\ &= 2\cos^2\theta - 1 \\ &= 1 - 2\sin^2\theta\end{aligned} \qquad \sin 2\theta = 2\sin\theta\cos\theta \qquad \tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

$$\cos^2\theta = \frac{1+\cos 2\theta}{2} \qquad \sin^2\theta = \frac{1-\cos 2\theta}{2} \qquad \tan^2\theta = \frac{1-\cos 2\theta}{1+\cos 2\theta}$$

$$\sin A\sin B = \frac{1}{2}\big(\cos(A-B) - \cos(A+B)\big) \qquad \sin(A+B) = \sin A\cos B + \cos A\sin B$$

$$\cos A\cos B = \frac{1}{2}\big(\cos(A-B) + \cos(A+B)\big) \qquad \cos(A+B) = \cos A\cos B - \sin A\sin B$$

$$\sin A\cos B = \frac{1}{2}\big(\sin(A+B) + \sin(A-B)\big) \qquad \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A\tan B}$$

$$c^2 = a^2 + b^2 - 2ab\cos C \qquad \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

You may use the rest of this formula sheet page for rough work. It will not be graded. If you need more space for a solution, please use one of the blank pages at the back of the test.

1. For each of the following questions, only your final answer will be graded.

No justification is necessary.

(1a) (1 point) Compute $\lim_{x\to-\infty} \frac{\cancel{x^3 - \sqrt{4x^6+2}}\ 2x^3}{x^2+3x^3-2x}$.

Final Answer

1

$\frac{3x^3}{3x^3}$

(1b) (1 point) Let $f(x) = x^2 \ln(e^{2x} + 1)$ and calculate $f'(x)$.

Final Answer

$$2x\ln(e^{2x}+1) + \frac{2x^2e^{2x}}{(e^{2x}+1)}$$

$$2x\ln(e^{2x}+1) + \frac{x^2}{(e^{2x}+1)} \times (2e^{2x})$$

(1c) (1 point) Let $f(x) = \arctan(\sin(x) + 3x + 1)$ and calculate $f'(0)$.

Final Answer

2

$$\frac{4}{2} = \frac{1}{1+(\sin(0)+0+1)^2} \times (\cos(0)+3)$$

(1d) (1 point) Estimate $e^{0.01}$ by using a linear approximation.

Final Answer

1.01

$\frac{d}{dx} = e^{a} = e^{a}$

$a=0$ $1\ e^{0} + e^{a}(x-a)$

$1+x = 1+0.01 = 1.01$

$3 \times 2 \times \frac{1}{3} = 2$ $g(4) = (f^{-1}(4)+1)^2$

For each of the following questions, only your final answer will be graded.

$f^{-1}(4) = 2$

No justification is necessary.

$2(f^{-1}(\cancel{x}^{4})+1) \times \left(\frac{1}{f'(f^{-1}(4}\right.$

2. (2a) (2 points) Let $f(x)$ be an one-to-one function on $\mathbb{R}$ and define its inverse to be f^{-1}. We know the tangent line of $f(x)$ at 2 is $y = 3x - 2$. Let $g(x) = (f^{-1}(x) + 1)^2$. Find $g(4)$ and $g'(4)$.

$f(2) = 4$

$f'(2) = 3$

Final Answer

$g(4) =$ 9

Final Answer

$g'(4) =$ 2

$2(\cos(x))^2 \times -\sin x$

$\frac{d}{x}(\cos^2(x)$

(2b) (2 points) Let $f(x) = (\cos^2(x) + 1)^x$. Compute $f'(x)$.

$\ln y = x \ln(\cos^2(x)+1)$

$2\cos(x) \times -\sin(x)$

$\frac{y'}{y} = \ln(\cos^2(x)+1) + x \frac{1}{(\cos^2(x)+1)}$

Final Answer

$$f'(x) = (\cos^2(x)+1)^x\left(\ln(\cos^2(x)+1) - \frac{2x\cos(x)\sin(x)}{\cos^2(x)+1}\right)$$

$10\frac{dy}{dx} = -10$

(2c) (2 points) A curve satisfies the equation $2x^2y = y^3 + x^2 + 3$. Define a function $y = f(x)$ near $(1, -2)$. Find the equation of the tangent line of $f(x)$ at $(1, -2)$.

-8 2

$4xy + 2x^2\frac{dy}{dx} = 3y^2\frac{dy}{dx} + 2x$

$y = f(1) + f'(1)(x-1)$

-2 -1

Final Answer

$y = -x - 1$

$4(-2) - 2 = -10 = 12 - 2$ $10\frac{dy}{dx}$ $\frac{dy}{dx} = -1$

(2d) (2 points) State the Intermediate Value Theorem. Make sure to specify all assumptions.

Final Answer

Let $a, b \in \mathbb{R}$

if $f(x)$ is continuous on $[a, b]$ and

$f(x)$ is differentiable on (a, b),

For $\forall v \in (f(a), f(b))$, $\exists c \in (a, b)$ s.t. $f(c) = v$

For each of the following questions, only your final answer will be graded.

No justification is necessary.

3. For each of the statement below, construct one function f with domain $\mathbb{R}$ such that f satisfies the given statement.

(3a) (2 points) f is (strictly) increasing on $\mathbb{R}$ and $f'(2) = 0$.

$y=(x-2)^3$

Final Answer

$y=(x-2)^3$

$f(0)=1$

(3b) (2 points) f is continuous on $\mathbb{R}$, f has a local max at $x = 0$, and $f'(0)$ does not exist.

$f(1)=2$

Final Answer

$y=-|x|$

$f(f(0))$

$f(f(x))$

$f(x)$ $f'(x)$

$y=x^2$ $f'(x)=|x|$ $f'(1)=f'(-1)=1$

$f'(f(0))f'(0) \leftarrow$ 1

$f'(0)=0$

(3c) (2 points) f is differentiable at 0 but $f \circ f$ is not differentiable at 0.

Final Answer

$$y=\begin{cases} x+1 & \text{for } x \neq 1 \\ |x|+1 & \text{for } x = 1 \end{cases}$$

$f'(1)$ DNE

$f'(0)$ exists BUT

$f'(f(0)) \leftarrow$ doesn't exist

x^n $\sqrt{x}$ $\frac{1}{x}$

sin/cos/tan

4. Let $f(x) = \sin x$.

(4a) (1 point) Find the largest interval containing 3 such that f is invertible on this interval.

$-\frac{\pi}{2}, \frac{\pi}{2}$ $\frac{\pi}{2}, \frac{3\pi}{2}$ $\sin(\frac{3\pi}{2}) = -1$ $\sin(\pi) = 0$

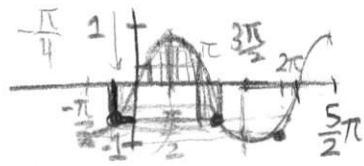
Final Answer

$\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$

(4b) (1 point) Define h to be the inverse function of f restricted on this interval. Which is the domain of this inverse function h.

$\frac{5}{4}\pi$ f D: $[\frac{\pi}{2}, \frac{3\pi}{2}]$

$\frac{3\pi}{2} - \frac{\pi}{2} = 2\pi$ R: $[-1, 1]$



Final Answer

$[-1, 1]$

(4c) (1 point) Find $f(h(0.6))$.

$f(f^{-1}(0.6)) = 0.6$

Final Answer

0.6

(4d) (1 point) Find $h(f(6))$. ?

$6 - \frac{3\pi}{2} = \frac{3\pi}{2} - t$ 6 $t = 3\pi - 6$

Final Answer

$3\pi - 6$

(4e) (1 point) Find h in terms of arcsin.

$h = -\arcsin(x)$

$-\sin(x) = f(x)$ (in terms of values)

$\sin(-\frac{\pi}{2}) = -1$ $f(\frac{\pi}{2}) = 1$

arcsin $-\frac{\pi}{2} + \pi$

Final Answer

$-(\arcsin(x) - \pi)$

(4f) (2 points) Sketch $h(f(x))$ on $[-3\pi/2, 3\pi/2]$. Make sure to label your axes and that you have the correct domain.

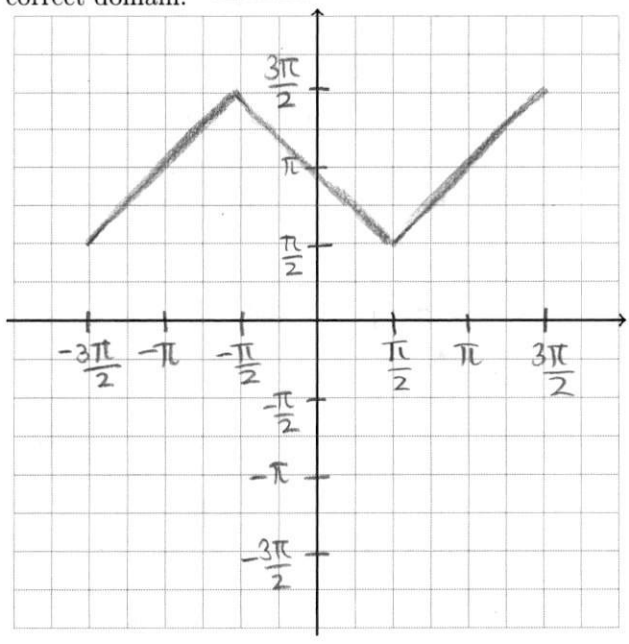
$\arcsin(-1) = -\frac{\pi}{2}$

$h(-1) = \frac{3\pi}{2}$

$\arcsin(1) = \frac{\pi}{2}$

~~arcsin(~~

$h(1) = \frac{\pi}{2}$



$h(f(x)) = x$

D: $[-\frac{3\pi}{2}, \frac{3\pi}{2}]$

$f(x)$ is one to one for $[\frac{\pi}{2}, \frac{3\pi}{2}]$

R of $h(x) = [\frac{\pi}{2}, \frac{3\pi}{2}]$

5. (5a) (1 point) Let $a \in \mathbb{R}$. Write the limit definition of "f is differentiable at a".

$$\lim_{x\to a} f'(x) = \lim_{x\to a} \frac{f(x)-f(a)}{x-a} \text{ exists for } a$$

(5b) (6 points) Assume that $a > 0$. Let f be a function defined on $\mathbb{R}$ that is differentiable at a. Define g by $g(x) = \dfrac{f(x)}{\sqrt{x}}$ for all $x > 0$. Prove that g is differentiable at a and that

$$g'(a) = \frac{\sqrt{a}f'(a) - \frac{f(a)}{2\sqrt{a}}}{a}.$$

Write a proof directly from the definition of the derivative as a limit without using any of the other differentiation rules, e.g. quotient rule or chain rule. No epsilon-delta proof is needed for this question. *Hint: you may need to use* $x - a = (\sqrt{x} - \sqrt{a})(\sqrt{x} + \sqrt{a})$ *in your proof.*

Let f be a function defined on $\mathbb{R}$ that is differentiable at a. By the definition of the derivative, for $x = a$,

$$\lim_{x\to a} g'(x) = \lim_{x\to a} \frac{g(x)-g(a)}{x-a} = \lim_{x\to a} \frac{\frac{f(x)}{\sqrt{x}} - \frac{f(a)}{\sqrt{a}}}{x-a} = \lim_{x\to a} \frac{\sqrt{a}f(x) - \sqrt{x}f(a)}{\sqrt{ax}\,(x-a)}$$

$$= \lim_{x\to a} \frac{\sqrt{a}f(x) - \sqrt{a}f(a) + \sqrt{a}f(a) - \sqrt{x}f(a)}{\sqrt{ax}\,(x-a)} = \lim_{x\to a} \frac{\sqrt{a}(f(x)-f(a))}{\sqrt{ax}\,(x-a)} +$$

$$\lim_{x\to a} \frac{(\sqrt{a}-\sqrt{x})f(a)}{\sqrt{ax}\,(\sqrt{x}-\sqrt{a})(\sqrt{x}+\sqrt{a})} = \lim_{x\to a} \frac{\sqrt{a}f'(x)}{\sqrt{ax}} + \lim_{x\to a} \frac{-f(a)}{\sqrt{ax}(\sqrt{x}+\sqrt{a})}$$

$$= \frac{\sqrt{a}f'(a)}{\sqrt{a\times a}} + \frac{-f(a)}{\sqrt{a\times a}(\sqrt{a}+\sqrt{a})} = \frac{\sqrt{a}f'(a) - \frac{f(a)}{2\sqrt{a}}}{a}$$

f is differentiable (and thus continuous) at a so $f'(a)$ and $f(a$ exists, and given $a > 0$ as $g(x) = \frac{f(x)}{\sqrt{x}}$ for all $x > 0$, g is differentiable at a and $g'(a) = \frac{\sqrt{a}f'(a) - \frac{f(a)}{2\sqrt{a}}}{a}$

6. (5 points) Let $f(x) = \frac{1}{3+x^4}$. We say that a rectangle R is inscribed between the graph of f and the x-axis, if two top vertices of R are on the graph of f and two bottom vertices are on the x-axis. What is the largest possible area of a rectangle inscribed between the graph of f and the x-axis? Show all of your work including justifying your answer is the largest possible area.

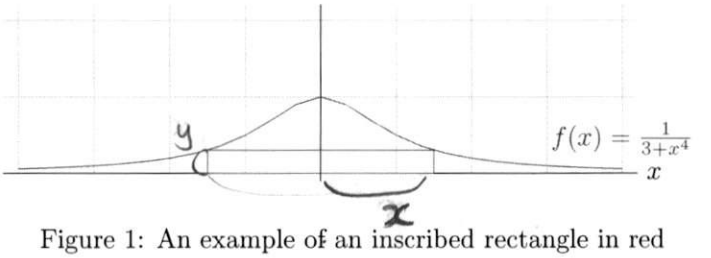


Figure 1: An example of an inscribed rectangle in red

$2 \times 2 = 4$

$\frac{1}{2}$ $3 + \cancel{2^4}$ $16+3$

$1 \times \frac{1}{3+(\frac{1}{2})}$

Area of $R = 2xy$

$f(x) = y = \frac{1}{3+x^4}$ $f(-x) = y = \frac{1}{3+(-x)^4} = \frac{1}{3+x^4}$

Area of $R = 2xy = \frac{2x}{3+x^4} = r(x)$

To find the largest possible area, we have to find critical points of the function $r(x)$ and choose the x value that results in the greatest area, $r(x)$.

$$\frac{d}{dx} r(x) = \frac{\frac{d}{dx}(2x) \times (3+x^4) - \frac{d}{dx}(3+x^4) \times (2x)}{(3+x^4)^2} = \frac{2(3+x^4) - 4x^3 \times 2x}{(3+x^4)^2}$$

$$= \frac{6+2x^4-8x^4}{(3+x^4)^2} = \frac{6-6x^4}{(3+x^4)^2} = \frac{6(1-x^4)}{3+x^4}$$

$r'(x) = 0$ when $x = \pm 1$

the side of a rectangle cannot be negative so $x = 1$

$$r(x) = 2xy = \frac{2x}{3+x^4} = \frac{2}{3+1} = \frac{2}{4} = \frac{1}{2}$$

$\therefore \frac{1}{2}$ is the largest possible area of R.

7. (7a) (2 points) State the Mean Value Theorem.

Let $f(x)$ be a function defined on interval I and $a, b \in I$

If $f(x)$ is continuous on $[a,b]$ &

$f(x)$ is differentiable on (a,b),

$\exists c \in [a,b]$ s.t. $f'(c) = \dfrac{f(b)-f(a)}{b-a}$

(7b) (5 points) Consider the function $f(x) = \ln(x)$ on the interval $[1/3, 3]$. Prove that

$$\forall x_1, x_2 \in [1/3, 3],\ |f(x_1) - f(x_2)| \leq 3|x_1 - x_2|$$

Let $f(x) = \ln(x)$ on $[\frac{1}{3}, 3]$

$f(x)$ is an increasing function for the interval $[\frac{1}{3}, 3]$ so

Case 1: $x_1 > x_2$:

- since $x_1 > x_2$, $f(x_1) > f(x_2)$

$|f(x_1) - f(x_2)| = f(x_1) - f(x_2) \leq 3(x_1 - x_2)$

According to the MVT, $\exists c \in [x_2, x_1]$ s.t. $f'(c) = \dfrac{f(x_1) - f(x_2)}{x_1 - x_2}$

$f'(x) = \dfrac{1}{x}$ and the largest value of $f'(x)$ is 3 since

$f'(\frac{1}{3})$ is $1 \div \frac{1}{3} = 3$ and $\frac{1}{x}$ is a decreasing function for $x \in [\frac{1}{3}, 3]$

Since $c \in [x_1, x_2]$ and $x_1, x_2 \in [\frac{1}{3}, 3]$, $f'(c) = \dfrac{f(x_1) - f(x_2)}{x_1 - x_2} \leq 3$.

This can be transformed to $f(x_1) - f(x_2) \leq 3(x_1 - x_2)$, so this inequality is true for $x_1 > x_2$.

Case 2: $x_1 = x_2$

$\ln(x) = f(x)$ is an injective function, so $f(x_1) = f(x_2)$.

$|f(x_1) - f(x_2)| = 0 \leq 0 = 3(0) = 3|x_1 - x_2|$

so this inequality is true for $x_1 = x_2$.

☆ Case 3 is on the next page

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Case 3: $x_1 < x_2$

$$|f(x_1) - f(x_2)| = f(x_2) - f(x_1) \leq 3(x_2 - x_1) = 3|x_1 - x_2|$$

This inequality is true for $x_2 > x_1$ the same way Case 1 is proven to be true, the only difference is that x_2 is now the bigger of the two values.

As this inequality was proven true for all 3 cases,

$$\forall x_1, x_2 \in [\tfrac{1}{3}, 3],\ |f(x_1) - f(x_2)| \leq 3|x_1 - x_2|$$